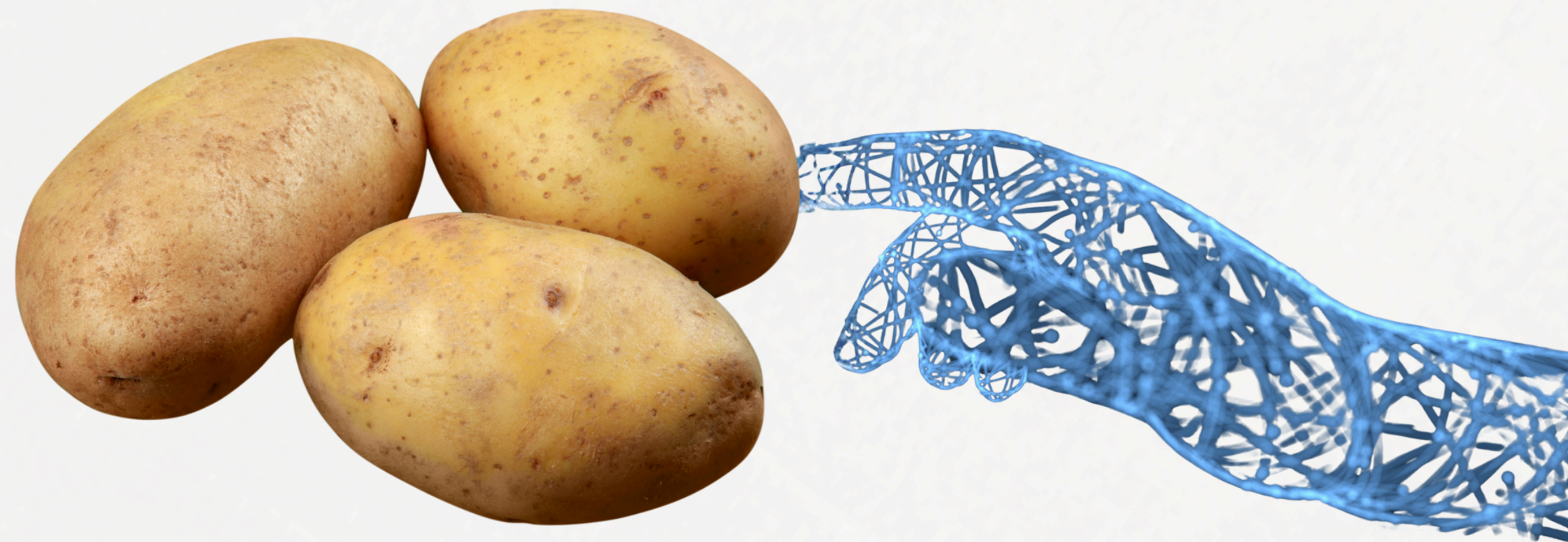




Using AI to analyse your data

HADC! (HARPER ADAMS DATA CLUB)

Ed Harris

2025-10-21

— Problem



You want to analyse data
and communicate it

Problem



Software



You want to analyse data
and communicate it



Data

$$y = mx + b$$

Stats knowledge



What you want to learn

Problem



Software

Terror zone



You want to analyse data
and communicate it

$$y = mx + b$$

Stats knowledge



Data

Comfort zone



What you want to learn

Problem



Statistical consultant

Terror zone



Software

$$y = mx + b$$

Stats knowledge



You want to analyse data
and communicate it



Analysis plan



What you want to learn

Comfort zone



Experimental
design



Data



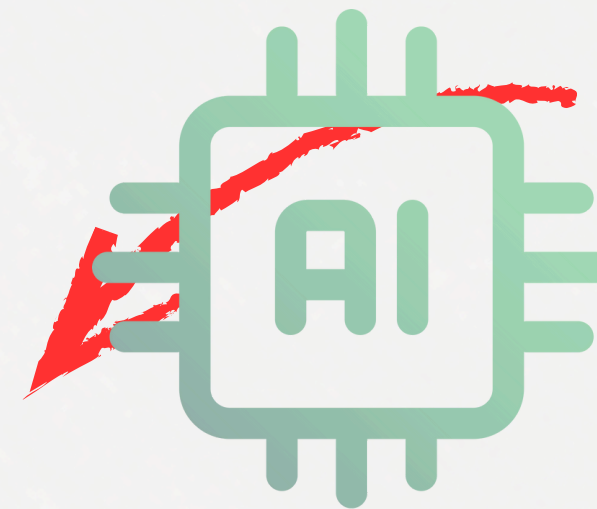
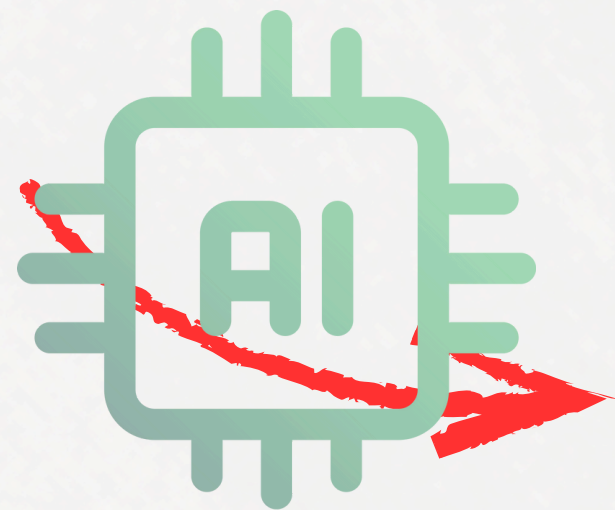
Tidy data

Solution

New comfort zone?



Software



$$y = mx + b$$

Stats knowledge

You want to analyse data
and communicate it



Data

Comfort zone



What you want to learn

—
Solution

How?

AI bros...

Lots of options...

What is best practice?

—
Solution

AI versus Ed



—

Demo 1

Ed's analysis

Demo 2

AI “informed” analysis

—

Demo 3

AI naive analysis

Discussion

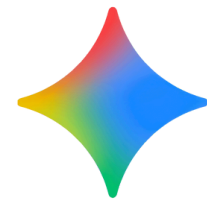
AI Platforms, options



ChatGPT



Copilot



Gemini



Grok



GitHub
Copilot



Claude
Sonnet 4.5



CURSOR



NotebookLM



Studio®

—
Discussion

Prompting and context

(Are these real skills?)

Platform	“Advanced” stats (GLM mixed models)	Details (ignores advanced stuff like agents, MCP, etc.)
Cursor (Alternatives: Positron, RStudio+Claude code or copy/paste, etc., account costs money)	 can produce and run code with output	-Possibly the best, most popular platform for AI coding -Limited free mode (somewhat stingy) -Requires setup, runs things locally (on your computer) -Access to best/many LLM models from different companies -Requires some skills; you control everything
ChatGPT Pro (Advanced Data Analysis mode, pro account costs money)	 can produce and run code with reasonable output	-Costs money -Works pretty well, potential to get better -Requires some skills; you control everything -Some ability to run things remotely
Windsurf (Decent free mode , for now...)	 similar to cursor	
ChatGPT (vanilla mode free mode ; alternatives: Claude Sonnet, Google Gemini, etc.)	 can produce good code	-Pretty good at R and python code -Can’t actually run the code or directly edit files -You need to be able to articulate exactly what you want
NotebookLM (free mode)		
JuliusAI (free-ish mode)	 simple stats/regression only	-free mode stingy -promises a lot but only delivers in simple stats -not suitable for real research (black box for methods, reproducibility, etc.)